LSA Types in OSPF

LSAs are packets that contain detailed information about routers, links, and routes within a network. Each router generates LSAs and shares them with other routers. These LSAs are then combined to create a complete map of the network, which is used to calculate the shortest path to different destinations using the Shortest Path First (SPF) algorithm.

OSPF networks can become large and complex, especially when divided into multiple areas. To manage this complexity, different types of LSAs are used. Each type has a specific role, such as describing routers, networks, or external routes. This classification helps OSPF remain organized, efficient and scalable.

Main LSA Types in OSPF

1. Type 1 - Router LSA

The Router LSA is generated by every router in the network. It contains information about the router's interface and its connection to neighboring routers. This LSA is only shared within the same OSPF area.

This is the most fundamental LSA type and it is essential for building the network topology.

2. Type 2 - Network LSA

The network LSA is generated by the Designated Router (DR) in a multi-access network, such as ethernet. It describes all routers connected to that network segment.

This LSA helps reduce unnecessary complexity by centralizing information about shared networks

3. Types 3 - Summary LSA

Summary LSAs are generated by Area Border Routers (ABRs). Their purpose is to advertise routes from one OSPF area to another.

They enable communication between different areas and help maintain a hierarchical network structure.

4. Type 4 - ASBR Summary LSA

This LSA is also generated by ABRs. It provides information about how to reach an Autonomous System Boundary Router (ASBR).
In simple terms, it acts as a pointer that guides routers towards the source of external routes.

5. Type 5 - External LSA

External LSAs are generated by ASBRs. They are used to advertise routes that originate the ~~OSP~~ outside the OSPF network, such as routes from other routing protocols.
This allows OSPF to connect with external networks and share those routes internally.

Additional LSA Types:

Type 6 - Group Membership LSA

This type is rarely used and is associated with multicast OSPF.
It is generally not important for beginner-level understanding

Type 7 - NSSA External LSA

This type are used in Not-So-Stubby Areas (NSSA). They are similar to Type 5 LSAs but are restricted within the NSSA. These LSAs are later converted into Type 5 LSAs by an ABR when leaving the NSSA.